Quality Management

9 — Camera sensors: from photons to grey levels

Prof. Gianluca Rossi • **dr. Lorenzo Capponi**
BSc in Engineering Management • *Università degli Studi di Perugia*
A.Y. 2026/2027 • 19 October 2026



Today

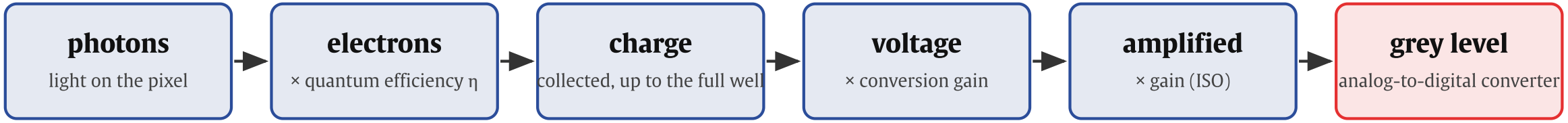
The link between the lens and the numbers

Lectures 7 and 8: the lens forms an image, and the image becomes a matrix of numbers. Today: the **sensor** that does the conversion — how it works, and what limits every measurement made with it.

1. **Photons to electrons**
   photodiode, pixel, quantum efficiency
2. **Pixels to an image**
   CCD and CMOS, shutters, line and area
3. **Colour, spectrum, size**
   Bayer filter, IR, sensor formats
4. **What limits it**
   noise, saturation, dynamic range
5. **Choosing a camera**
   for the washers, and for a line

The sensor is a measurement chain

Five conversions between the light and the grey level



- **Quantum efficiency η** — electrons generated per photon arriving, between 0 and 1. Typical: 0.5–0.8 in green light.
- **Full-well capacity** — the most electrons a pixel can hold before it saturates. About 10 000 for a 3.45 µm pixel.
- **Conversion gain** — volts per electron, then grey levels (DN, digital numbers) per electron after the converter.

One standard to compare cameras

EMVA 1288 defines how machine-vision cameras are measured and what their datasheets must state: quantum efficiency, read noise, full well, dark current, non-uniformity. Two cameras compared on EMVA data are compared fairly.

PART 1

From photons to an image

Photodiode · Pixel · CCD and CMOS · Shutters · Line and area · Colour · Spectrum · Size

The photodiode

Every pixel is one

- A photon absorbed in silicon frees an **electron** — if its energy exceeds silicon's band gap, 1.12 eV. That sets the longest wavelength silicon can see: about **1.1 µm**.
- The electrons are collected and counted. The current is proportional to the light:

$$I = \eta \cdot q \cdot P / (h\nu) \qquad R = I / P$$

P: optical power · $h\nu$: energy of one photon · q: charge of the electron · R: responsivity, in A/W.

Proportional to the light is what makes a photodiode a **measuring** element — the same reason it is used in light meters, barcode readers and fibre-optic probes.

HOW MANY ELECTRONS

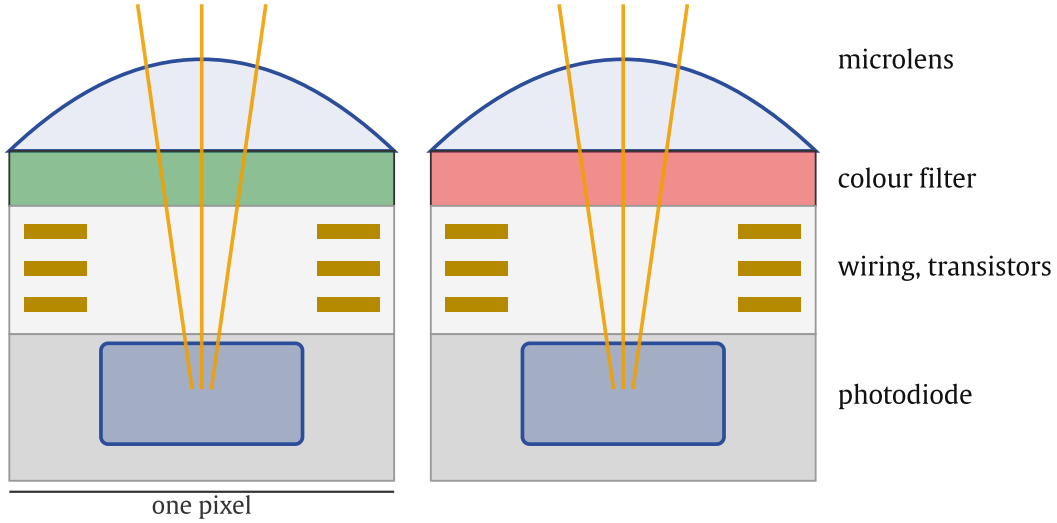
Green light, 550 nm: one photon carries 3.6×10^{-19} J. A power of 1 nW is 2.8×10^{9} photons per second. With $\eta = 0.6$: 1.7×10^{9} electrons per second, a current of **0.27 nA**, and a responsivity of 0.27 A/W.

Counting, not measuring a voltage

A camera pixel does not measure a current: it **collects** electrons during the exposure and counts them at the end. A pixel's signal is a number of electrons.

Inside a pixel

A photodiode, a filter and a lens, a few micrometres wide



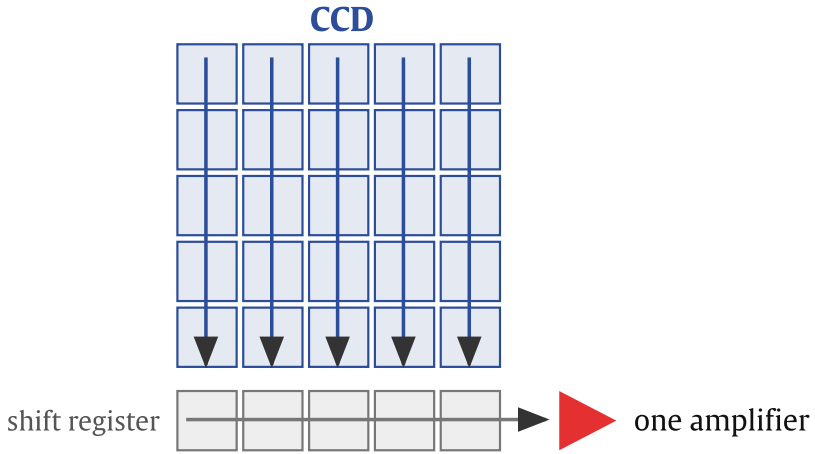
- **Microlens:** focuses the light onto the photodiode, so light falling on the wiring is not lost.
- **Colour filter:** on colour sensors only — one colour per pixel (slide 10).
- **Photodiode:** collects the electrons during the exposure. The fraction of the pixel's area that is sensitive is the **fill factor**.
- **Wiring and transistors:** read the charge out. In *back-side illuminated* sensors they sit below the photodiode, so they no longer shade it.

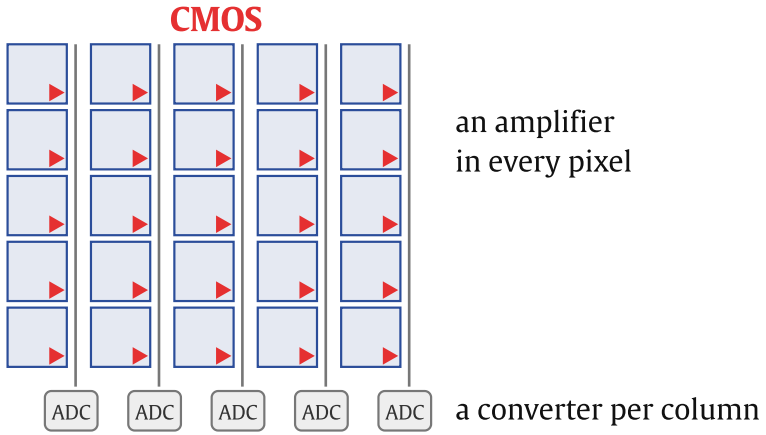
Pixel size

Phones: 0.6–1.4 µm. Machine vision: 2–6 µm. Larger pixels collect more light and hold more electrons: less noise (Part 2).

Reading the pixels: CCD and CMOS

The difference is where the charge becomes a voltage





CCD — charge-coupled device

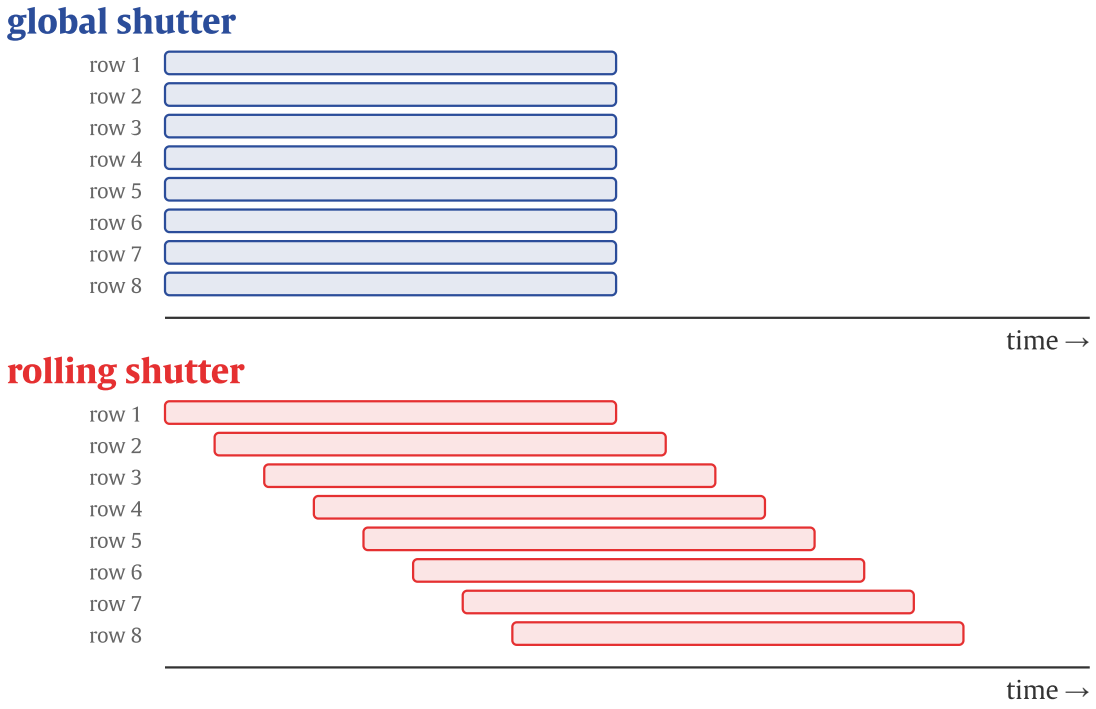
The charges are shifted, row by row, to **one** output amplifier. Every pixel is read by the same electronics: very uniform and low-noise, but slow and power-hungry. Production has largely ended.

CMOS — complementary metal-oxide semiconductor

Every pixel has its **own amplifier**, and each column its own converter. Fast, low power, cheap — and it can read just a **region of interest**: half the rows, about twice the frame rate. Today almost every industrial camera is CMOS.

Global and rolling shutter

Do all rows look at the scene at the same moment?



- **Global shutter:** all rows start and stop collecting light together. A moving part is recorded as it was at one instant.
- **Rolling shutter:** each row starts a little after the one above. Cheaper and common in CMOS — and in every phone. A moving part is **skewed**.

ON A CONVEYOR

Reading 2048 rows takes 20 ms. A part moving at 0.5 m/s travels **10 mm** between the first row and the last: a round washer comes out as an ellipse.

For moving parts

A global shutter, or a rolling shutter with a **flash** fired while all rows collect. For still washers on a light box, a phone's rolling shutter is fine.

Area-scan and line-scan cameras

A grid of pixels, or one line and a moving part

Area-scan
A two-dimensional grid: the whole field in one image. For discrete parts — washers, castings, assemblies.

Line-scan
One row of pixels (sometimes a few). The image is built line by line **as the part moves past**, triggered by an encoder on the line. For continuous material — paper, sheet steel, textiles, glass, film — and for cylinders rotated in front of the camera.

INSPECTING A STEEL STRIP

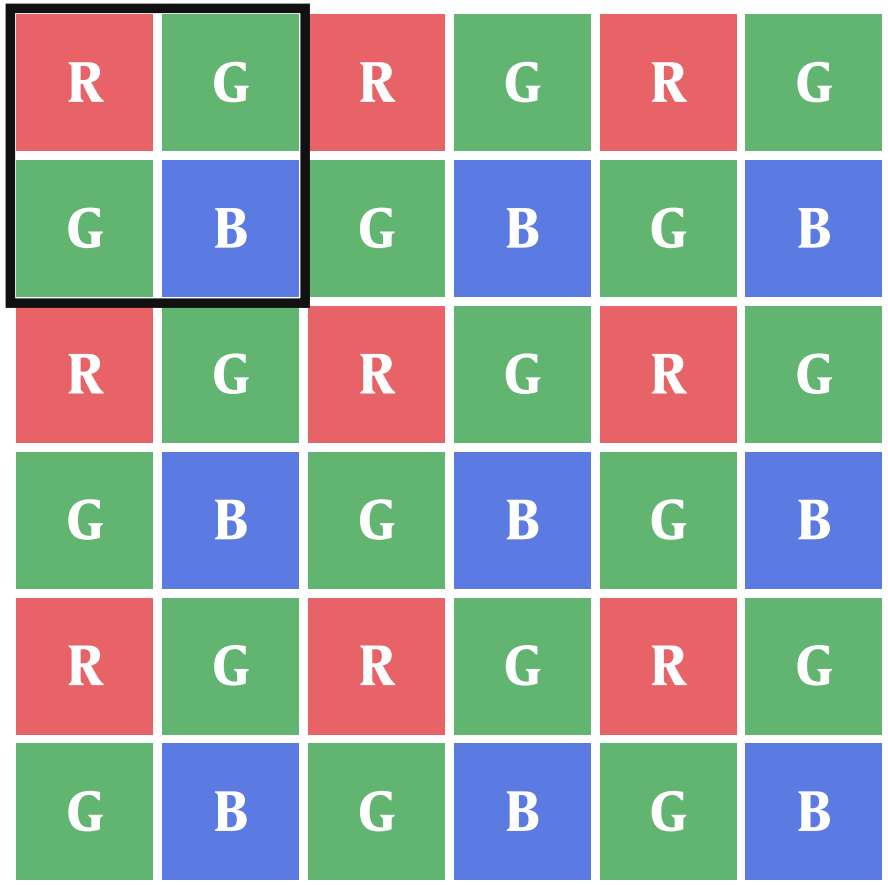
A strip 1 m wide, one line of 8192 pixels: **0.12 mm** per pixel across. The strip moves at 2 m/s; for square pixels the camera must take a line every 0.12 mm — **16 400 lines per second**.

Across the strip the resolution is set by the pixels; along it, by the speed and the line rate. If the speed changes and the line rate does not, the image stretches — which is why the trigger comes from an encoder, not a clock.

TDI (time-delay integration) line cameras have several rows and add them up as the part moves past: more light per line on fast lines, with the same line lights.

Colour sensors

A photodiode does not see colour: a filter in front of it does



half the pixels green, a quarter red, a quarter blue

- **Bayer filter:** a mosaic of red, green and blue filters, one per pixel, in the pattern R G / G B. Green appears twice because the eye is most sensitive to it — it carries most of the brightness.
- **Demosaicing:** each pixel records one colour; the other two are **estimated from its neighbours**. Edges pick up colour fringes, and the real resolution of each colour is lower than the pixel count.
- **Three-chip cameras:** a prism splits the light onto three sensors, one per colour — full resolution in each, but expensive and delicate to align.

For measurement: monochrome

No filters: every pixel records the full signal, with two to three times more light. Gauging cameras are monochrome; your phone is a Bayer camera (lab exercise 1).

What the sensor can see

Beyond the visible — and why that is useful on a line

Range	Sensor	Use in quality control
Visible, 400–700 nm	silicon, with an **IR-cut filter**	what people see: colour, marking, assembly
Near infrared, 700–1000 nm	silicon, **without** the IR-cut filter	lighting invisible to operators, and insensitive to the room's light
Short-wave IR, 0.9–1.7 µm	InGaAs	moisture in food and wood, sorting plastics, seeing through silicon wafers
Thermal IR, 8–14 µm	microbolometer	heat, not light: welds, electrical faults, insulation, hidden defects
Ultraviolet	UV-enhanced silicon	fluorescent markings, cracks shown by penetrant inspection, coatings

The IR-cut filter

Silicon sees up to 1.1 µm, the eye only to 0.7. Without the filter, a colour camera mixes infrared into its colours and the image looks wrong: consumer cameras always have one.

Choose the light first

The sensor's range only matters together with the light: a near-infrared LED and a filter on the lens make an inspection that ignores the sunlight through the factory windows.

Sensor size and format

The names are historical; the dimensions matter

Format	Diagonal	Typical use	Crop factor
1/2″	8 mm	machine vision, phones	5.4
2/3″	11 mm	machine vision	3.9
1″	16 mm	machine vision, compacts	2.7
APS-C	about 28 mm	cameras	about 1.5
Full frame	43.3 mm (36 × 24)	cameras	1

"1/2 inch" is not a dimension of the sensor: it comes from the diameter of the old television tubes that sensors replaced. Read the size in millimetres. **Crop factor** = 43.3 mm / diagonal: the same lens on a smaller sensor sees a narrower field.

THE WASHER CAMERA

2448 × 2048 pixels of 3.45 µm: 8.45 × 7.07 mm, diagonal 11.0 mm — a **2/3″** sensor.

The lens must cover the sensor

Every lens projects a circle of light. If it is smaller than the sensor's diagonal, the corners go dark. Machine-vision lenses are sold by format: a 2/3″ lens for a 2/3″ sensor.

More pixels on the same area means smaller pixels: less light each, fewer electrons, more noise.

Recap TAKEAWAYS

Part 1 — from photons to an image

1. **The chain:** photons → electrons (quantum efficiency) → charge (up to the full well) → voltage → grey levels. EMVA 1288 is how cameras are compared.
2. **Photodiode:** current proportional to light, up to about 1.1 µm in silicon. A pixel collects and counts electrons during the exposure.
3. **CCD vs CMOS:** one amplifier for all, or one per pixel. CMOS is fast, reads regions of interest, and is what industry uses today.
4. **Moving parts:** global shutter or a flash; continuous material: a line-scan camera triggered by an encoder.
5. **Colour, spectrum, size:** Bayer cameras estimate two colours out of three; measure in monochrome. Choose the spectral range with the light. Read sensor sizes in millimetres.

PART 2

What limits the sensor

Noise · Signal-to-noise ratio · Calibration · Saturation · Dynamic range · Linearity

Where the noise comes from

The same scene never gives exactly the same numbers twice

Source	Cause	Size	What reduces it
Shot noise	light arrives as photons, at random	$\sqrt{N}$ for N electrons	more light — nothing else
Dark current	electrons freed by heat, not light	grows with exposure time; doubles every 6–8 °C	short exposures, cooling
Read noise	the amplifier and converter	1–10 electrons per read	a better camera
Quantisation	rounding to grey levels	$\Delta/\sqrt{12}$	more bits
Fixed-pattern noise	every pixel slightly different in offset and gain	the same in every image	**calibration** (slide 17)

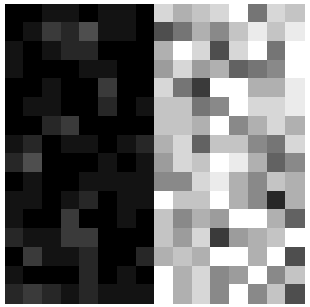
The first four change from image to image: they are **random** and average out over many pixels or many images. The last one is **systematic**: it does not average out, but because it repeats, it can be measured once and subtracted.

Signal-to-noise ratio

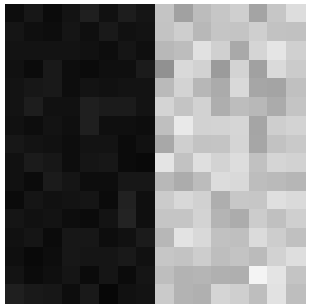
Shot noise sets the limit: four times the light for twice the quality

$$\mathrm{SNR} = N / \sqrt{(N + \sigma_{\mathrm{read}}{}^2 + \sigma_{\mathrm{dark}}{}^2)}$$

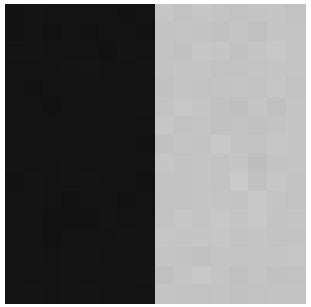
In good light shot noise dominates and **SNR ≈ $\sqrt{N}$**. A pixel holding 10 000 electrons can reach SNR = 100 at most; doubling the SNR takes four times the light.



10 electrons
SNR ≈ 3



100 electrons
SNR ≈ 10



10 000 electrons
SNR ≈ 100

THE WASHERS, READ NOISE 5 ELECTRONS

Background at 80 % of the full well: 8000 electrons, SNR ≈ 89. Washer at 8 %: 800 electrons, SNR ≈ 28. Both far above what a threshold needs.

Averaging

Averaging k independent values divides random noise by $\sqrt{k}$ — over pixels, or over images. It is why fitting hundreds of edge points beats a pixel (lecture 12), and why you take several photos in the lab.

Gain does not help

Raising the gain (ISO) multiplies signal and noise together: the SNR stays the same, and the brightest values saturate sooner. More light is the only real fix.

Fixed-pattern noise and its calibration

Measure the camera's own errors once, then remove them from every image

- **Offset differences** between pixels (dark-signal non-uniformity), and a few **hot pixels** that read bright in the dark.
- **Gain differences** (photo-response non-uniformity): under the same light, pixels give slightly different values. Lens vignetting adds to it.

The correction

Dark frame D: images with the lens covered. **Flat field** F: images of a uniform white surface. Then every image I is corrected as

$$I_c = (I - D) \,/\, (F - D) \cdot \mathrm{mean}(F - D)$$

Why it matters for inspection

A gauge that finds edges with a fixed threshold finds them in the wrong place where the image is darker. After flat-field correction, the same edge gives the same grey levels anywhere in the field.

Same idea as any instrument

Zero and span: the dark frame is the camera's **zero**, the flat field its **span**, pixel by pixel. Like any calibration, it is repeated when conditions change — temperature, exposure time, lens.

Saturation and blooming

When a pixel is full, the information is gone — and can spill over

- **Saturation:** a pixel that reaches its full well stops counting. Every brighter value reads the same: the detail in that region is lost, not just compressed.
- **Blooming:** in a CCD, the extra charge spills into the neighbouring pixels along the column — a bright streak. Anti-blooming drains and CMOS sensors largely prevent it.

SHINY PARTS

A machined metal surface reflects the lamp straight into the camera. The reflection saturates, a scratch inside it disappears, and the edge next to it moves outwards.

What to do

Diffuse light (a dome or a light tent), a **polarising** filter on the light and on the lens, or a shorter exposure. Check the histogram of every new setup: no peak at the top value.

Dynamic range

The brightest and the darkest thing one image can hold

$$\text{DR} = \text{full well} \,/\, \text{read noise}$$

$$\text{DR}_{\text{dB}} = 20 \log_{10}(\text{full well} \,/\, \text{read noise})$$

The ratio between the largest signal and the smallest one above the noise floor. In bits: $\log_2(\text{DR})$ — the number of bits worth recording.

THE WASHER CAMERA

10 000 electrons / 5 electrons = 2000: **66 dB**, about **11 bits**. A 12-bit converter records all of it; 8 bits throw some of it away — which is fine when, as on the light box, the scene needs far less.

When the scene exceeds it

A shiny part next to a black rubber seal can need more range than the sensor has. Then: change the lighting first; otherwise combine several exposures into one (HDR).

Linear cameras and processed images

A grey level is a measurement only if it is proportional to the light

Machine-vision camera

The output is **linear**: twice the light, twice the grey level. Every value can be traced back to a number of electrons.

Phone and consumer camera

Applies a **tone curve** (gamma, about value $\propto \text{light}^{1/2.2}$) so the picture looks natural, then sharpening, noise reduction, HDR and compression. The result looks better and measures worse: grey levels are no longer proportional to the light, and edges are altered.

For measurement

Use a linear camera, or the phone's **raw** format; switch off automatic exposure, white balance and HDR so every image is taken the same way.

In the lab you use a phone anyway — and part of the exercise is finding out how much of the result is the phone's processing rather than the washer. A threshold-based gauge is fairly robust to a tone curve; a measurement of brightness is not.

Recap TAKEAWAYS

Part 2 — what limits the sensor

1. **Noise:** shot, dark, read and quantisation noise are random; fixed-pattern noise is systematic. Random noise averages out, systematic noise is calibrated out.
2. **SNR ≈ $\sqrt{N}$ in good light:** four times the light for twice the SNR. Gain does not improve it.
3. **Calibration:** a dark frame and a flat field are the camera's zero and span, pixel by pixel.
4. **Saturation loses information for good:** diffuse the light, polarise, shorten the exposure; check that no peak sits at the top value.
5. **Dynamic range** = full well / read noise. Measure with linear output or raw images, never with the processed picture.

Choosing a camera for inspection

Two cases, the same questions

Question	Washers on a light box	Parts on a conveyor at 0.5 m/s
Area or line?	area-scan	area-scan, triggered by a sensor on the belt
Pixels and format	2448 × 2048, 3.45 µm, 2/3″	from field of view and smallest defect (lectures 7 and 8)
Shutter	rolling is fine: the parts are still	**global**, or a flash
Exposure	long enough to fill the well	under 82 µs, so a strong **strobe**
Colour?	monochrome	monochrome, unless colour is what is inspected
Spectrum and light	visible backlight	near-IR light and filter, to ignore the room
Bit depth	8 bits: the contrast is huge	10–12 bits if shiny and dark parts share the image
Image output	linear or raw, lossless	linear, region of interest for speed
Compare on	the **EMVA 1288** datasheet: quantum efficiency, read noise, full well, dark current, non-uniformity	

Next
Light and illumination: the part of the system that decides what the camera can see at all.

The hands-on lab
Your phone as the camera: its channels, its noise and its processing, then a gauge, then a decision on a batch.

Questions?

Prof. Gianluca Rossi • dr. Lorenzo Capponi

lorenzo.capponi@collaboratori.unipg.it • lorenzocapponi@outlook.it

github.com/LolloCappo/student-hub